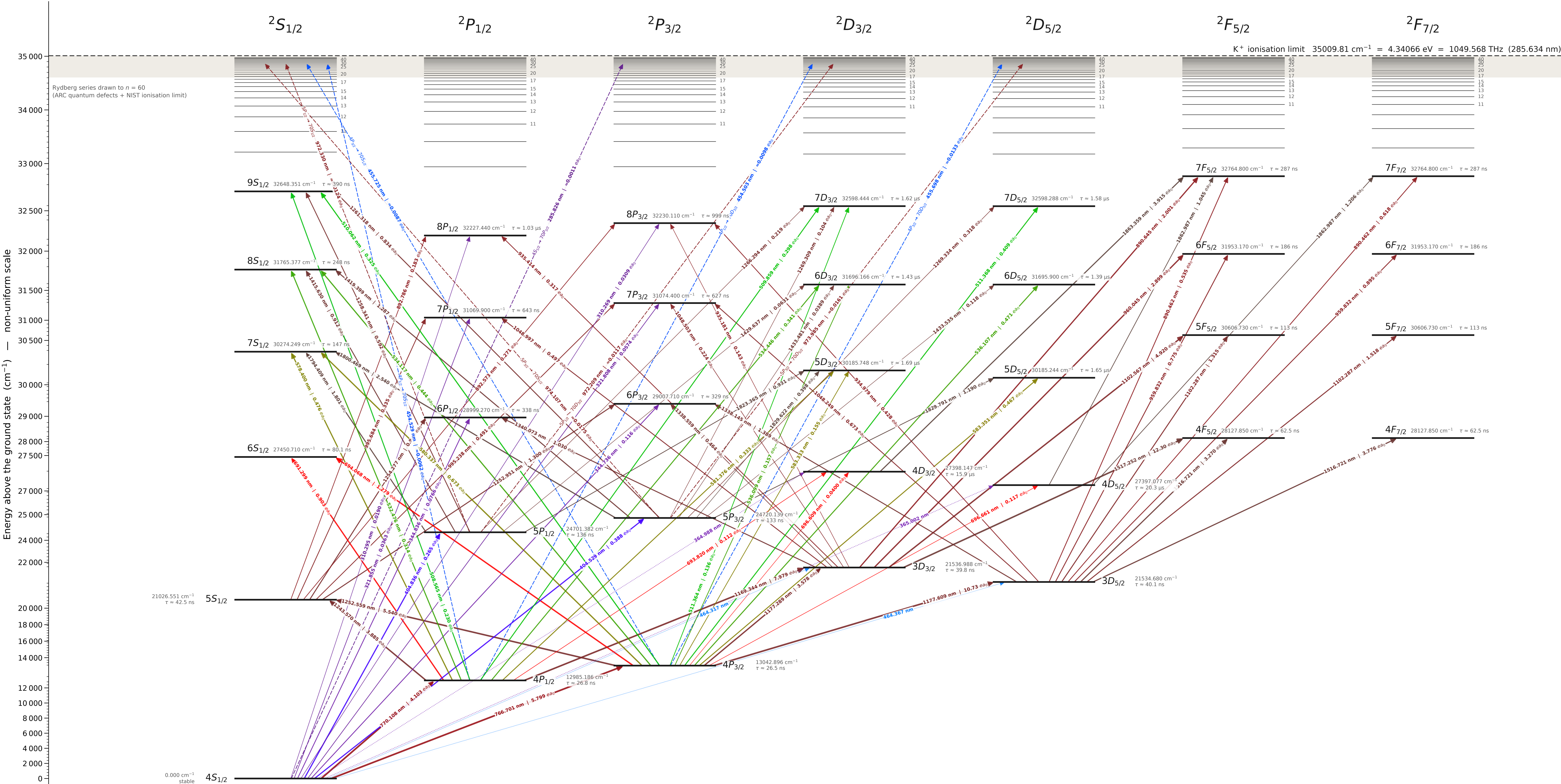
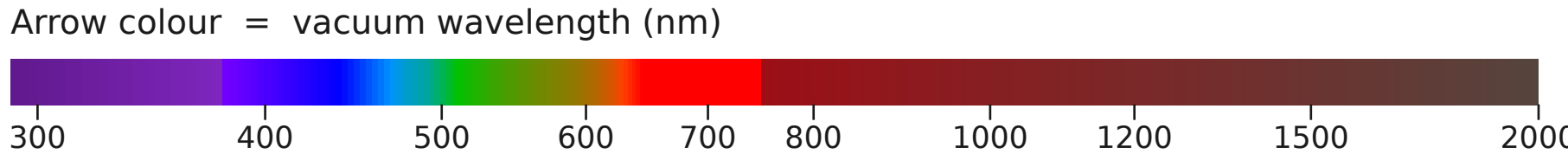


Potassium-39

³⁹K (K I), nuclear spin $I = 3/2$. Energy levels and transitions below 2 μm : 34 levels, 84 lines.

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J||er||J' \rangle$ in ea0
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and lifetime (measured where a value exists, otherwise ARC, marked $\approx$).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\tilde{\nu}$ (cm ⁻¹)	$\langle J er J' \rangle$ (ea0)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Typical use
4S _{1/2} → 4P _{3/2}	766.7009	766.4899	391.0162	13042.896	5.799	3.779e+07	6.006	100.0	
4S _{1/2} → 4P _{1/2}	770.1084	769.8964	389.2861	12985.186	4.103	3.734e+07	5.930	100.0	
4P _{3/2} → 3D _{5/2}	1177.6089	1177.2866	254.5773	8491.784	10.73	2.383e+07	3.973	100.0	
4P _{1/2} → 3D _{3/2}	1169.3442	1169.0241	256.3766	8551.802	7.979	2.017e+07	4.003	83.5	
4P _{3/2} → 5S _{1/2}	1252.5591	1252.2165	239.3440	7983.655	5.540	1.582e+07	3.744	66.5	
4P _{1/2} → 5S _{1/2}	1243.5699	1243.2298	241.0741	8041.365	3.885	7.951e+06	3.744	33.5	
4P _{3/2} → 6S _{1/2}	694.0678	693.8764	431.9354	14407.814	1.279	4.956e+06	1.987	40.5	
4P _{3/2} → 3D _{3/2}	1177.2889	1176.9667	254.6465	8494.092	3.578	3.974e+06	4.003	16.5	
4P _{1/2} → 6S _{1/2}	691.2988	691.1081	433.6655	14465.525	0.903	2.500e+06	1.987	20.4	
4P _{3/2} → 7S _{1/2}	580.3375	580.1766	516.5830	17231.353	0.673	2.348e+06	1.082	35.2	
4P _{3/2} → 8S _{1/2}	534.1173	533.9687	561.2858	18722.481	0.444	1.311e+06	0.641	33.1	
4P _{1/2} → 7S _{1/2}	578.4003	578.2400	518.3131	17289.063	0.476	1.186e+06	1.082	17.8	
4S _{1/2} → 5P _{3/2}	404.5285	404.4142	741.0911	24720.139	0.388	1.150e+06	1.198	16.8	
4S _{1/2} → 5P _{1/2}	404.8356	404.7213	740.5288	24701.382	0.265	1.070e+06	1.171	15.7	
4P _{3/2} → 9S _{1/2}	510.0621	509.9200	587.7568	19605.455	0.325	8.064e+05	0.408	32.0	
4P _{1/2} → 8S _{1/2}	532.4759	532.3279	563.0160	18780.191	0.314	6.620e+05	0.641	16.7	
4P _{3/2} → 6D _{5/2}	536.1067	535.9576	559.2030	18653.004	0.471	4.860e+05	0.114	56.7	

λ and ν from NIST level energies (centre of gravity, no hyperfine structure); λ air uses standard air (Ciddor).
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$. Branch: share of the upper level's decays that takes this line.

Rydberg excitation wavelengths (vacuum, nm)

Lower → series	n = 30	n = 40	n = 50	n = 70	n = 100	n → ∞	d ₇₀ ARC
4S _{1/2} → nP _{3/2}	286.757	286.246	286.019	285.826	285.727	285.634	0.0011
4P _{3/2} → nS _{1/2}	458.187	456.825	456.227	455.725	455.468	455.230	0.0087
4P _{3/2} → nD _{5/2}	457.819	456.676	456.152	455.698	455.459	455.230	0.0133
5S _{1/2} → nP _{3/2}	722.223	718.989	717.556	716.346	715.722	715.141	0.0039
3D _{5/2} → nP _{3/2}	749.737	746.253	744.708	743.406	742.734	742.108	0.0079
3D _{5/2} → nF _{7/2}	748.888	745.906	744.534	743.343	742.713	742.108	0.0555
5P _{3/2} → nS _{1/2}	985.427	979.149	976.402	974.107	972.932	971.848	0.0175
5P _{3/2} → nD _{5/2}	983.722	978.461	976.058	973.985	972.891	971.848	0.0161
4D _{5/2} → nP _{3/2}	1337.683	1326.632	1321.759	1317.661	1315.551	1313.588	0.0205
4D _{5/2} → nF _{7/2}	1334.983	1325.536	1321.209	1317.465	1315.485	1313.588	0.0898
6S _{1/2} → nP _{3/2}	1347.350	1336.139	1331.196	1327.039	1324.899	1322.908	0.0083
4F _{7/2} → nD _{5/2}	1479.780	1467.907	1462.506	1457.855	1455.407	1453.074	0.0176
6P _{3/2} → nS _{1/2}	1706.393	1687.655	1679.510	1672.731	1669.272	1666.082	0.0287
6P _{3/2} → nD _{5/2}	1701.288	1685.613	1678.495	1672.372	1669.151	1666.082	0.0173

Rydberg term energies from ARC quantum defects + NIST ionisation limit, lower level from NIST. d₇₀ = $\langle J||er||J' \rangle$ to n = 70 in ea0 (calculated).
S → nP values come from a near-cancelling integral: treat them as an order of magnitude.

³⁹K hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
4S _{1/2}	230.860	-	F=1: -288.57 F=2: +173.15
4P _{1/2}	28.850	-	F=1: -36.06 F=2: +21.64
4P _{3/2}	6.060	2.830	F=0: -19.19 F=1: -15.96 F=2: -6.67 F=3: +14.34
5S _{1/2}	55.500	-	F=1: -69.38 F=2: +41.62
5P _{1/2}	9.020	-	F=1: -11.28 F=2: +6.76
5P _{3/2}	1.969	0.870	F=0: -6.30 F=1: -5.20 F=2: -2.13 F=3: +4.65
6S _{1/2}	21.810	-	F=1: -27.26 F=2: +16.36
6P _{3/2}	0.886	0.370	F=0: -2.86 F=1: -2.34 F=2: -0.94 F=3: +2.09

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).